

Multi-Sensor Placement and Information Fusion Analysis to Enable Beyond Visual Line of Sight Operations for Small Uncrewed Aerial Vehicles

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Abstract—This paper describes the development of a multi-sensor placement and information fusion testbed for detection and tracking of small uncrewed aerial vehicles (sUAV). This testbed is called Mason's UAV Systems and Cyber Analysis Testbed (MUSCAT) and combines JDL Data Fusion model theoretic foundations with systems engineering principles applied to sUAV corridors design. Recent results are focused on analyzing sensor coverage with terrain models as a function of drone altitude, with a goal to identify optimal placement of multiple sensors and provide cost-performance trade-offs. The methodology is composed of four main phases: setting up the computational environment, defining sensor attributes and region of interest, computing sensor coverage for individual sensor types and respective combinations, and displaying results. An instance considering a drone corridor is used to analyze the optimal placement of short-range radars and remote identification (RID) receivers to maximize the total coverage and improve the system's resilience by track fusion with varied sensors. Results show that MUSCAT is able to generate a wide range of metrics and spatial insights, important to identify the most suitable solution based on user requirements.

Index Terms—Small UAVs, multi-sensors, optimal placement, drone corridors.

I. INTRODUCTION

Novel design and engineering concepts for small Uncrewed Aerial Vehicles (sUAV), commonly known as drones, are being introduced by multiple startups and established aerospace manufacturers alike. sUAVs offer capabilities ranging from operating air taxis and small cargo services within densely populated urban areas to enabling delivery of critical supplies (e.g., medical and rescue equipment) in difficult-to-reach rural and disaster-struck areas [1]. However, one of the

major challenges in the widespread adoption of drone operations—beyond sUAV manufacturers—is to enable Beyond Visual Line of Sight (BVLOS) operations. Nevertheless, given the solid economic growth that sUAV operations bring to a region, many federal and local state governments worldwide have recognized an eminent need to address this challenge. As a result, commercially viable operational concepts of drone superhighways and corridors are emerging in the United States (Rome, NY to Syracuse, NY), United Kingdom (Reading, Oxford, Milton Keynes, Cambridge, Coventry, and Rugby), and India (Delhi to Jhajjar), to name a few.

In these so-called drone corridors, consistent detection and continuous tracking of sUAV remains a prerequisite to ensure the safety and security of drone operations. Here a single sensor, such as RADAR, is ineffective because detection coverage is highly dependent on the RCS (which is already low for sUAV), terrain which remains unique to a region, and flight altitude that must be below general air traffic. Hence, such corridors deem a multi-sensor solution that includes radio frequency, passive systems, optical (visual and infrared bands), and acoustic sensors desirable.

This paper details implementing a multi-sensor placement and information fusion solution for detecting and tracking sUAV with an application in an under-development corridor in the Commonwealth of Virginia. Our research team is developing a drone operations analysis testbed, Mason's UAV Systems and Cyber Analysis Testbed (MUSCAT) [2], sponsored by the Commonwealth. This testbed combines JDL Data Fusion model [3] theoretic foundations with systems engineering principles applied to information system [4] for sUAV corridor analysis to guide multi-sensor acquisition decisions. Recent results are focused on analyzing sensor coverage with terrain models as a function of drone altitude to identify the optimal

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TABLE I
ADVANTAGES AND DRAWBACKS OF SENSOR TYPES TO DETECT sUAVS

Sensor type	Advantages	Drawbacks
Radar	Detects cooperative and non-cooperative targets; all weather	High acquisition price; short-range
RID	Receives flight information broadcasted by sUAVs	Dependent on emissions (cooperative targets)
RF Passive	Detects and identify RF emissions	Dependent on drones emissions
Acoustic	Detects and identify acoustic emissions	Short-range detection; environmental noise
Optical – visible	Detects light reflected by sUAVs	Short-range; weather conditions
Optical – IR	identification possible by shape Detects IR emissions by drones; identification possible by shape	Short-range; weather conditions

placement of multiple sensors and provide a cost-performance trade-off with sensor type, quality, and range. These sensors can be applied to provide information in a specific area with sUAV activities as well as in urban air mobility (UAM) corridors [5] [6].

The choice of number, type, and location of sensors is a challenging systems engineering design problem since each sensor type has specific characteristics and inherent advantages and drawbacks, as summarized in Table I. For example, radars can operate in all weather conditions and detect cooperative and non-cooperative drones. A *cooperative* sUAV is considered an aircraft that behaves and complies to be detected and followed by unmanned traffic management (UTM). In contrast, a *non-cooperative* drone is a vehicle that behaves to avoid or mask its detection and tracking. In general, radars to detect small targets have short ranges [7] and tend to be more expensive than other sensor types to detect sUAVs. Remote Identification (RID) receivers are comparatively cheaper and able to receive flight information broadcast from the drone to the UTM in a similar way as the automatic dependent surveillance-broadcast (ADS-B) in the crewed air traffic control [8] [9]. RID transmitters, however, are not expected to be used by non-cooperative sUAVs. Radio frequencies (RF) passive systems detect, identify, and eventually locate electromagnetic emissions from the air vehicle [10] [11]. RF passive systems have greater detection ranges than radars. However, they remain dependent on the emissions characteristics of the target (i.e., effective radiated power - ERP) and the existence of line-of-sight (LoS) between the drone and the receiver. In other words, if the target does not transmit, the transmitted power is low, or the emission source is masked by terrain or distances involved, passive RF systems may be unable to provide detection. Acoustic and optical systems are comparatively cheaper than radars and RF passive sensors. However, they are highly affected by the environment (i.e., urban noise and weather) and have much shorter detection ranges [12].

One important reason to consider diverse types of sensors in our study is to analyze the resultant system resilience by sensors fusion [13] [14] [15] [16]. A single sensor option to detect and track sUAVs in a corridor may be less expensive. However, a multi-sensor coverage with track fusion can offer superior resilience; for example, in case of loss of one type

of sensor (i.e., by a cyber-attack), the remaining equipment shall be able to keep the detection and tracking in the affected region. In addition to different sensor characteristics, range, and prices, the choice of sensor placement location is not straightforward since the terrain affects the sensor coverage, as well as the installation and maintenance of equipment depend on geographical and legal constraints such as infrastructure (i.e., electric power) and environment (i.e., elevations, restricted areas).

Given the different characteristics of multiple sensors, in this paper, we discuss a system engineering trade-off analysis in MUSCAT by developing a modeling “sensor coverage cost per square mile” in a given drone corridor to assist decision-makers with infrastructure investment decision that leverage multi-sensor data and information fusion capabilities to ensure safety and security of sUAV operations. The remainder of the paper is dedicated to drone corridor optimization problem formulation and using computational means in MUSCAT to develop and analyze the sUAV corridor operations.

II. MULTI-SENSORS OPTIMIZATION PROBLEM

This section discusses our research question, which can be viewed as a resource allocation problem. In this type of problem, an optimal allocation of a fixed amount of resources to activities must be determined to minimize costs incurred by the allocation [17]. In this way, our question can be expressed as: *what is the best number, combination, and placement of sensors in a given region of interest to obtain the best system performance with limited resources?* A related research addressing network design and sensor placement for UAM can be found in [18]. In our case, we first determine the geographic sector in which the traffic of drones needs to be known. This *region of interest* (ROI) has size, shape, and elevations, and to be covered, sensors can be physically inside or outside the ROI. Only a set of limited locations can be used since the installation and maintenance of sensors are affected by many aspects such as electricity supply, type of space if public or private, nearby elevations, airfields, etc. The following mathematical model can express the analysis of the combination of different types of sensors to obtain the best coverage for a given target type.

Sets and Indices

- T Set of sensor types, indexed by $t \in T$.
- T' Subset of sensor types, indexed by $t' \in T'$.
- L Set of locations, indexed by $l \in L$.
- L' Subset of locations, indexed by $l' \in L'$.
- U Set of sUAVs, indexed by $u \in U$.
- H Subset of flying altitudes, indexed by $h \in H$.

The sensor set T is given by different types of sensors, i.e., active and passive, such as radar, passive radio frequency detectors and direction finders, optical (visual and IR), and acoustic. Sensor location L is given by a set of possible locations to install the sensor, either inside or outside the ROI. The sUAV type U is composed of the various characteristics of the vehicle, especially size and respective radar cross-section (RCS). The drone altitude is represented by the target altitude

H , directly impacting the line-of-sight (LoS) and the sensor coverage for the respective altitude. For simplicity, sensors' antenna heights are always considered 10m (33ft) over the terrain in our model.

Parameters	
AB	Available budget.
C_t	Costs of sensor type t .
SC_{tluh}	Coverage of sensor type t in location l detecting an sUAV type u flying at altitude h .
σ	A conversion parameter.

The available budget AB represents cost limits given by the decision maker that may provide some flexibility when considering subjective aspects such as benefits from an increase in resilience. The parameter C_t represents the cost of determined sensor type t . SC_{tluh} expresses the sensor coverage for a given sensor type t in the location l detecting a UAV u flying at altitude h . The conversion parameter σ calculates sensor coverage and costs in the Objective Function.

Decision variable

$$x_{tluh} = \begin{cases} 1 & \text{if sensor } t \text{ is used in location } l \text{ to detect target } u \text{ flying at altitude } h; \\ 0 & \text{otherwise.} \end{cases}$$

Objective function
Maximize:

$$\sum_{t \in T} \sum_{l \in L} \sum_{u \in U} \sum_{h \in H} SC_{tluh} \times x_{tluh} - \sigma \times \sum_{t \in T} C_t \times x_{tluh} \quad (1)$$

Subject to:

$$\sum_{t \in T} C_t \times x_{tluh} \leq AB \quad \forall l \in L, u \in U, h \in H \quad (2)$$

$$\sum_{u \in U} \sum_{h \in H} x_{t'l'uh} = 1 \quad \forall t' \in T', l' \in l' \quad (3)$$

$$\sum_{u \in U} \sum_{h \in H} x_{t'l'uh} = 0 \quad \forall t' \in T', l' \in l' \quad (4)$$

$$x_{tluh} \in \{0, 1\} \quad \forall t \in T, l \in L, u \in U, h \in H \quad (5)$$

In the above model, the objective function Eq(1) represents the ROI covered by sensors. Eq(2) states that the costs of sensors do not exceed the available budget. Eq(3) allows the model to assign sensor type t' to specific locations l' . Eq(4) prevents the model from assigning sensor type t' to specific locations l' . Eq(5) states as binary the decision variable x_{tluh} . For example, Eq(3) could be used to guarantee that at least one radar will be placed near an airport for safety reasons. Similarly, Eq(4) could be applied to avoid acoustic sensors being located next to highways. One important part of the solution methodology involves calculating sensor coverage since other results, such as total costs, depend on it. A similar but reduced situation can be found in the example "Planning Radar Network Coverage over Terrain" [19]. In summary, this example considers a single type of sensor (radar) that can be

placed in six locations of a given region of interest (ROI) to detect aerial targets. Locations are near mountains, and respective terrain elevation data must be considered propagation effects to calculate the signal-to-noise ratio (SNR) obtained from a target in different parts of the ROI. This region is rectangular, defined by two geographic coordinates, and divided into small sectors or cells where detection is considered if the SNR exceeds a certain threshold. As a final result, we have a 3D map with the best coverage of three radars. The same principle can be used as part of our solution methodology. However, our problem requests specific features such as:

- Coverage of different types of sensors.
- Resultant costs.
- ROI of irregular shapes.

The multi-sensor coverage (i.e., radars and RID) not only gives the idea of the total area covered and the contribution of each sensor but also where this coverage is simultaneous. It is important to understand where tracking fusion could be obtained. Also, it provides data for cost analysis considering the coverage and coverage gaps in a given ROI with a varied number and type of sensors. The ROI is significant because it limits the region where the SNR and other parameters are calculated to indicate detections. In our case, the ROI can coincide with the sUAV corridor. These corridors may assume a complex route to link origin and destination instead a straight line [6]. Then, using a square or rectangular ROI may not be adequate. If the ROI is excessively large to accommodate a complex path, the simulation time can be significantly increased. The developed solution methodology is explained in the next section to address these and additional requirements.

III. METHODOLOGY FOR SENSOR PLACEMENT AND FUSION IN SUAV CORRIDORS

This section describes the methodology developed to optimize different sensors placement. This diversity is fundamental to obtaining data fusion from different sensors and increasing the system resilience, which is significant in case of cyberattacks.

A. Algorithm Development

The algorithm was designed to determine and optimally display sensor coverage within a specified ROI. The algorithmic approach can be structured into four core segments: setting up the computational environment, defining sensor attributes and ROI, computing sensor coverage for individual sensor types and their combinations, and displaying the results, discussed as follows.

1) *Setting up the environment:* The computational environment is initially prepared by integrating customized terrains into the algorithm. For sensor coverage, it is essential to consider terrain elevation data. For this study, we incorporated DTED maps from the US Geological Survey to perform subsequent computations. Simultaneously, a user interface object is created to facilitate the display of results later in the process.

2) *Defining sensor attributes and ROI*: In this stage, we thoroughly define the attributes and locations of each sensor type. We also specify the ROI by indicating its latitude and longitude boundaries. The user determines the ROI grid cell size as a desired resolution for the area covered. The meshing process may generate a new area different than the initial ROI, for example, if the shape of the area is irregular, as shown in Figure 1. This expansion occurs as the original area (a) is adjusted to accommodate the grid cells (b). Consequently, the resulting meshed area represents a sum of all the squares (c). Although the meshed area may surpass the initial ROI, it ensures a comprehensive coverage analysis by including potential edge cases that lie on the border of the ROI.

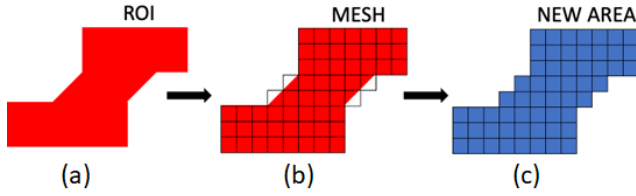


Fig. 1. Using a meshing process to create grid cells in a ROI of irregular shape.

3) *Computing sensor coverage*: Upon finalizing the sensor attributes and the ROI, sensor coverage over the ROI is computed for each sensor type. This procedure involves analyzing sensor-specific information in tandem with target positional data. Notably, the position of each grid cell acts as a prospective target, and detection is determined through a detection model equation unique to each sensor type. Acknowledging that different sensor types necessitate specific equations for modeling target detection is crucial. For instance, the radar equation is employed for radars; paired with the target's RCS, we can ascertain whether a target is detected by setting a certain detection threshold [20]. The RID is a communication link where the transmitter reaches the receivers with power equal to or superior to receiver sensitivity.

After completing the above segments, we calculate the coverage for each possible sensor position. The algorithm subsequently assesses the coverage of all potential combinations of sensors. Take a scenario involving three sensors — A, B, and C — the small triangles in Figure 2. The possible combinations are A, B, C, AB, AC, BC, and ABC. The algorithm then identifies the optimal coverage scenario for each combination of sensors. At this juncture, multi-sensor analysis is conducted, considering the best individual result of each sensor for a specific grid cell without accounting for potential enhancements from sensor fusion within the area. Upon completing computations for all sensor types, the algorithm consolidates these diverse data sets. Total coverage is obtained by integrating the coverage scenarios of different sensor types, presenting a comprehensive overview of sensor coverage across the ROI. This involves pairing the optimal coverage scenario for each quantity of a sensor type with the best scenario for the same quantity of other sensor types. A

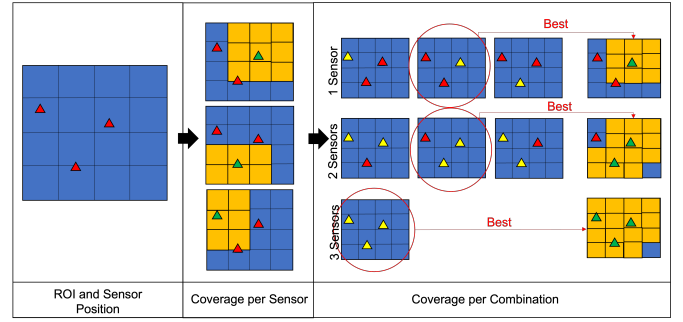


Fig. 2. The algorithm for optimal sensor coverage calculates and selects the best coverage for each sensor combination. Considering three potential sensor locations, each coverage is combined to determine the ideal coverage for different placement and number of sensors.

binary approach is adopted to simplify the complex task of integrating different sensor types and their associated detection models. This strategy transforms each sensor type's detection model output into a binary format 0/1, respectively, for non-detection and detection. Although this simplification may not account for potential enhancements from sensor fusion, it nonetheless provides a robust strategy for identifying optimal architectures during the early stages of a sensor placement project. Calculating total coverage is further depicted in Figure 3.

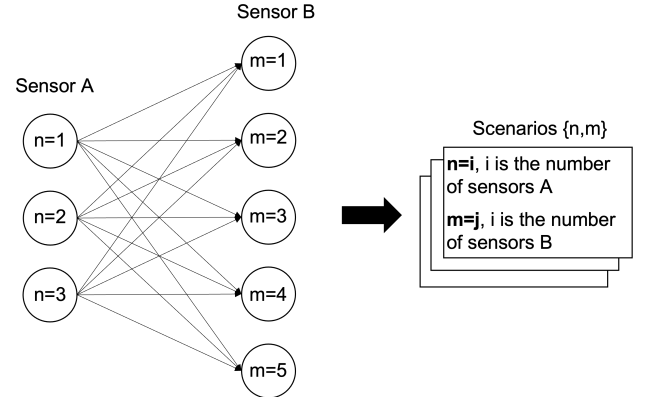


Fig. 3. Total coverage is derived by integrating the coverage scenarios of different sensor types.

B. Displaying results

The computed coverage is visually depicted using the previously established user interface object. To provide a comprehensive and intuitive understanding of the sensor coverage across the ROI, we employ maps to illustrate coverage areas and graphs to compare different scenarios encompassing various combinations of sensor types and quantities. This dual-method visualization not only makes the computation results accessible but also fosters a deeper understanding of the overall sensor distribution and performance within the ROI. The core data in MUSCAT is the coverage, which is represented by the index M_c as in Equation 6, where n is the

number of grid cells with coverage and N is the total number of grid cells.

$$M_c = n/N \quad (6)$$

This coverage data serves as the foundation for calculating other important metrics. For each sensor, metrics such as gaps (index M_g in Equation 7) can be calculated, providing insights into the effectiveness of the sensor placement.

$$M_g = 1 - n/N \quad (7)$$

The metric of overlap is another crucial aspect that MUSCAT considers. It is calculated by accumulating the information of sensor detections for each cell. This metric provides valuable insights into the redundancy and overlap of coverage between different sensor types, enabling a comprehensive assessment of the monitoring system.

Another essential metric is the cost per area coverage CA , which quantifies the economic efficiency of the sensor placement strategy. Equation 8 is utilized to calculate this metric and needs the cost of each sensor C_t , allowing designers to evaluate the cost-effectiveness of the coverage achieved in a specific area.

$$CA = \sum_{t \in T} C_t \times x_{tluh} / M_c \quad \forall t \in T, l \in L, u \in U, h \in H \quad (8)$$

To visualize the workflow of the testbed, Figure 4 provides a clear depiction. This flowchart showcases the steps involved in using MUSCAT for sensor coverage analysis, highlighting the input data, calculations, and resulting metrics.

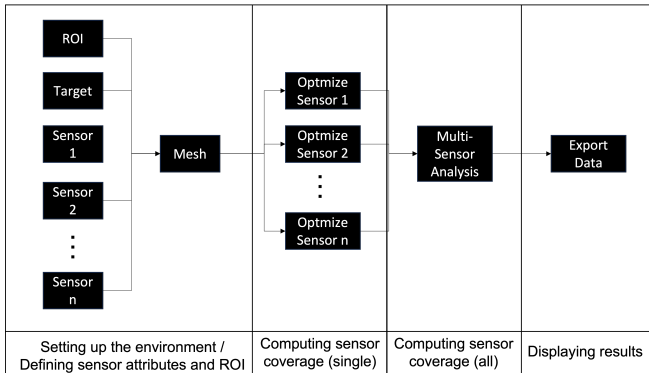


Fig. 4. Total coverage is derived by integrating the coverage scenarios of different sensor types.

In the next section, we present an instance considering a drone corridor to demonstrate the analysis of the optimal placement of short-range radars and remote identification (RID) receivers to maximize the total coverage and indicate simultaneous coverage, essential to verify where track fusion can be obtained.

IV. SIMULATION MODEL AND RESULTS ANALYSIS

A. Simulation Set-up

This simulation is considered a corridor between Stafford and Warrenton airports (Figure 5). In this case, we assume that

TABLE II
RADAR PARAMETERS USED IN THE SIMULATION.

Parameter	Value
Radar	
Frequency	18GHz
Antenna gain	38dBi
Probability of detection	0.8
Probability of false alarm	10^{-9}
Pulses integrated	10
Maximum range	5km
RID	
Frequency	2.4GHz
Sensitivity	-90dBm
Target	
RCS	$0.1m^2$
Altitude	400ft
ERP	10W

a straight path cannot be used (i.e., to avoid a sensitive area), and the ROI has an irregular shape. Cyan diamonds indicate the possible installation sites for RID sensors (eight in total), while red diamonds denote the possible locations for Radar sensors (eight in total). These sensors play a crucial role in the overall system. The diagram displays a white grid with square cells measuring 0.005 degrees on each side, representing the meshed ROI. This grid is a reference for precisely placing the sensors, ensuring optimal coverage and functionality.

It is also necessary to the simulation some parameters about the sensors, targets, ROI, and available locations to install the equipment. In this instance, we will consider combining one type of short-range radar and one type of RID receiver to obtain track fusion using two different sensors and improve the system's resilience. The sensors considered are a short-range monostatic radar and a RID. The parameters used in the simulation regarding the sensors and the target are in Table II.

The assessment of terrain elevation is conducted using a DTED model obtained from the US Geological Survey. This model provides accurate and detailed elevation information with a resolution of 1-arc second. The elevation data from the DTED model enables informed decisions and optimize the system design accordingly.

To illustrate the problem at hand, let us consider two specific requirements for the corridor:

- The coverage of the area should exceed 95%; and
- The overlap between areas covered by two different types of sensors must be greater than 55%.

The first requirement ensures a comprehensive monitoring solution. The second requirement is about redundancy. Besides that, we will focus on achieving the minimum viable architecture. Furthermore, we will introduce a cost for radars and RIDs to analyze the cost-effectiveness of the sensor deployment. For this example, let's assume a cost of 5 units of measure (UoM) for radars and 1 UoM for RIDs. Considering these requirements and costs, we can explore the trade-offs and make informed decisions regarding the corridor's sensor placement and system design.

Although these requirements are relatively simple, they serve as a practical example to showcase the utilization

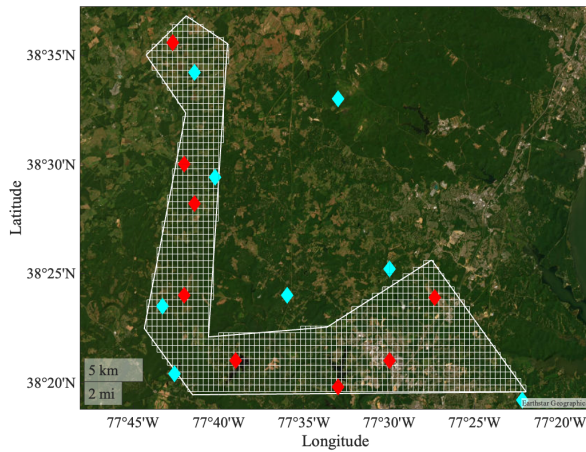


Fig. 5. The considered ROI is the irregular shape area inside the sUAV corridor. The cyan diamonds are available locations to install RID sensors, and the red diamonds are available to install Radar sensors.

of the MUSCAT methodology. The system systematically optimizes sensor placement, considering various factors such as coverage, overlap, and redundancy. With this development, we can efficiently determine the optimal placement of sensors in the corridor, ensuring the fulfillment of both requirements and maximizing the effectiveness of the monitoring system.

B. Results Analysis

After executing MUSCAT for the designated ROI depicted in Figure 5, the analysis yielded the coverage results for each type of sensor, as illustrated in Figure 6. MUSCAT's advanced algorithms and calculations precisely determined the extent of coverage achieved by each sensor type and the infrastructure needed to achieve the requirements.

The analysis of individual sensor coverage provides insights

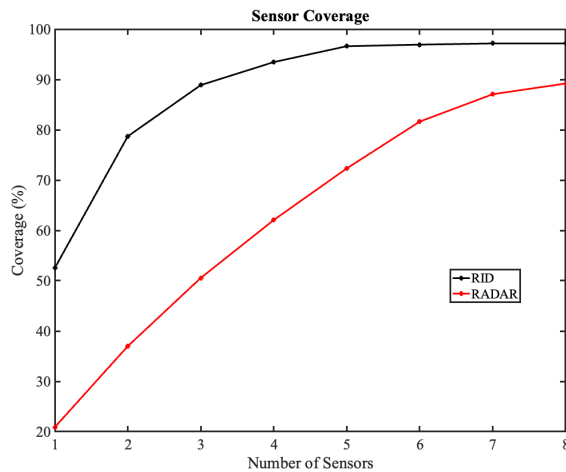


Fig. 6. Comprehensive coverage breakdown by sensor type.

into the specific roles played by each sensor within the corridor

(Figure 6). For the RID sensors, it was observed that the coverage reached a ceiling of approximately 97% with just five sensors, meeting the 95% coverage requirement. The increase in coverage from 1 to 3 RID sensors was found to be substantial, nearly doubling the coverage. However, the increase became minimal beyond three sensors, reaching a ceiling when the quantity is five RIDs. This information is vital for determining the minimum infrastructure needed for a corridor. It could also be utilized in negotiations with stakeholders, considering that adding more RID sensors beyond five would not significantly enhance the coverage (Figure 6).

On the other hand, the radar sensors exhibited different characteristics. Due to their smaller range and sparse distribution, the area was not saturated with radar coverage, and a ceiling was not reached (Figure 6). However, the graph showed that with the addition of the first four radar sensors, approximately 60% of the area was covered. The subsequent four radars only contributed around half of that coverage, resulting in 89%.

While the individual coverage of sensors could be assessed using Figure 6, it is crucial to consider the complementary effect between sensors to achieve higher coverage. For such cases, a graph like the one projected in Figure 7 becomes necessary. This type of graph depicts the coverage achieved by combining sensors, providing insights into how different sensor types work together to maximize coverage.

It is important to highlight that despite deploying eight RIDs

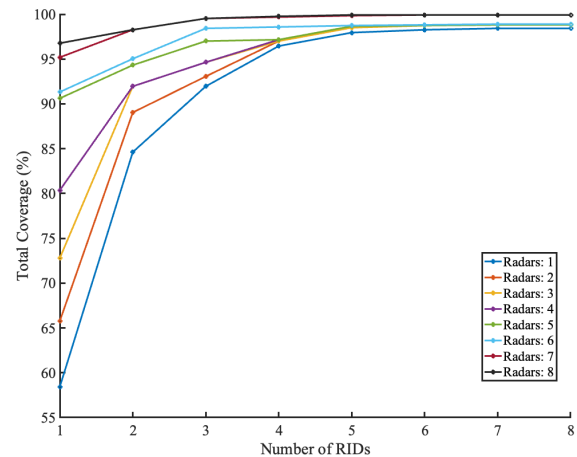


Fig. 7. Cumulative coverage by combined sensors.

and eight radars, achieving 100% coverage still needs to be attained. The maximum coverage obtained was 99.92%. If the requirement mandates full 100% coverage for the designated area, exploring alternative sensor placement positions would be possible. This underscores a valuable aspect of MUSCAT, as it assesses coverage and guides corridor designers in identifying additional sensor positions that may be required to fulfill the desired coverage objective.

The overlap of different sensors is another essential tool for determining the optimal combinations that satisfy the requirements, as shown by Figure 8. This graph focuses explicitly on

the overlap resulting from using different sensors, excluding any overlap that may occur from the same type of sensor. Figure 7 illustrates the cumulative coverage given sensor combinations, being possible to assess the alternatives that fulfill the requirements of coverage and redundancy also shown by Figure 9. These complementary visual representations provide valuable insights into identifying the sensor configurations that offer the desired coverage and overlap, aiding in the decision-making process for designing an effective sensor network.

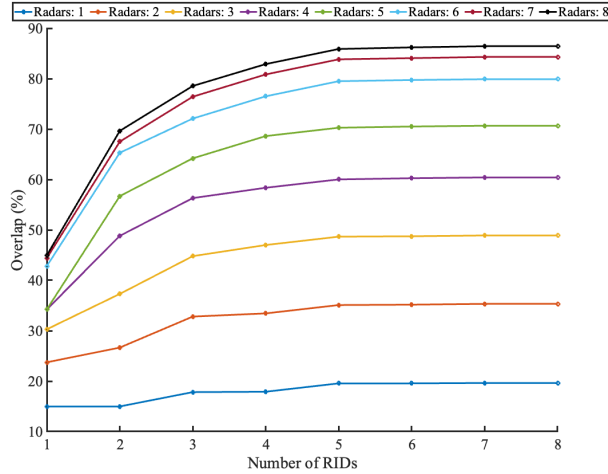


Fig. 8. Coverage breakdown by sensor type.

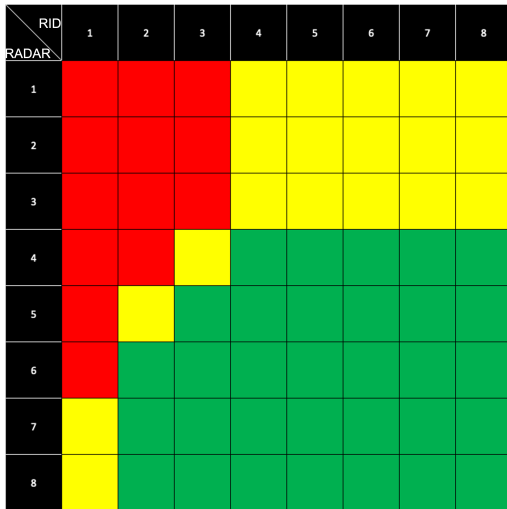


Fig. 9. Combinations of sensors, including their types and quantities, to check the achievement of requirements. In the legend, the green color represents combinations that fulfill both requirements simultaneously. Yellow represents combinations that only meet one requirement, while red represents combinations that do not fulfill any requirement.

We highlight the significance of MUSCAT in generating various metrics once it enables a comprehensive evaluation of different sensor combinations. For instance, considering five radars and two RIDs, it is observed that the requirement for overlap is met, but not the coverage.

TABLE III
COST PER COVERAGE AREA FOR THE THREE PROPOSED SOLUTIONS

Sensor combination	Cost (UoM)	Coverage (%)	Cost-effectiveness (UoM/%)
4 Radars + 4 RIDs	24	97.16	0.25
5 Radars + 3 RIDs	28	97.00	0.29
6 Radars + 2 RIDs	32	95.03	0.34

Looking closely at Figure 9, three potential candidates emerge as solutions for the problem as they balance effectiveness and cost. These candidates are four radars plus four RIDs, five radars plus three RIDs, and six radars plus two RIDs. These combinations present favorable outcomes compared to other options that incur higher costs.

To further refine the decision-making process, the final metric to consider is the cost per coverage area. This metric helps assess each candidate solution's cost-effectiveness, allowing for a more informed and economically viable choice. By analyzing this metric, a comprehensive understanding of the optimal sensor configuration can be achieved.

Table III presents the cost per coverage area for the three proposed solutions. These values are important to achieve an optimal balance between coverage requirements and cost-effectiveness. Among the configurations, the one with four radars and four RIDs stands out as it achieves the desired coverage while maintaining a reasonable cost per coverage area. By selecting this solution, effective corridor monitoring can be ensured while keeping expenses minimized. Another essential aspect to consider is the spatial distribution of coverage. In certain scenarios, specific requirements may demand obligatory coverage by a particular type of sensor in certain regions. For instance, near an airport, radar coverage might be necessary. MUSCAT is capable of providing such information, as exemplified in Figure 10.

This figure illustrates the coverage achieved by deploying four radars and four RIDs. The positions of each sensor are carefully selected to ensure maximum coverage for the entire set. This type of analysis is essential because the cluster of gaps in the southern part of the corridor could pose a problem if there are requirements that prohibit such gaps. This ensures that all critical areas receive adequate coverage based on specific requirements, enhancing the effectiveness and reliability of the surveillance system.

C. Discussion and Limitations

The utilization of MUSCAT in this study has provided valuable metrics and spatial distribution insights into sensor coverage, individually and when deployed jointly. Inherent capabilities extend to determining the optimal positions for a given number of sensors within a set of potential positions, facilitating effective sensor placement strategies.

One notable advantage of the developed system is its efficient and straightforward approach to comparing different types of sensors using the binary approach. This approach simplifies the comparison process, allowing designers to assess the coverage capabilities of different sensor types more

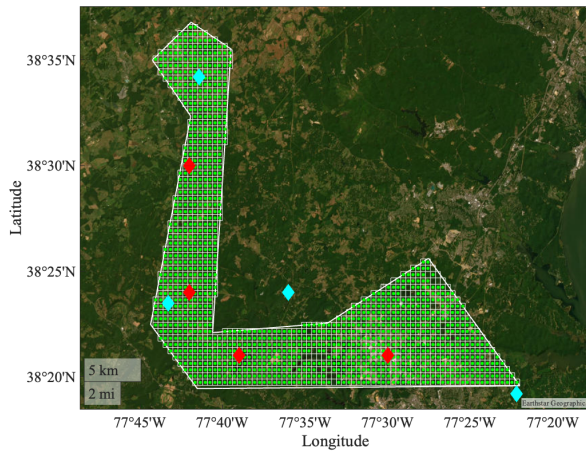


Fig. 10. Spatial analysis of coverage distribution for four radars and four RIDs. The legend color scheme illustrates grid cells with coverage from at least one sensor, represented by green dots.

practically and meaningfully. Another strength is its flexibility in usage throughout the various phases of corridor project development. The testbed can accommodate different models of sensors, making it applicable during both the early design and later stages involving specific sensor implementation. It is important to note that MUSCAT does not aim solely to find the definitive best solution. Instead, it provides designers with a wide range of metrics and spatial insights, empowering them to make informed decisions and identify the most suitable solution based on their specific requirements. The testbed, however, does have some limitations. As it works with a discretization of areas, the computation demands can increase if the analyzed area is highly refined and large. This may require substantial computational resources to evaluate each cell's coverage effectively. Furthermore, expanding the system to include additional sensor types requires implementing mathematical models specific to those sensors. These models provide information on whether detection occurs at a given point, enabling comprehensive coverage analysis.

V. SUMMARY AND FUTURE WORK

In summary, this paper presents the implementation of the MUSCAT as a multi-sensor placement and information fusion solution for detecting and tracking sUAS in a corridor under development in the Commonwealth of Virginia. The developed system is a valuable sensor placement and coverage analysis tool providing metrics and spatial insights aiding decision-makers by empowering designers with crucial information to optimize the effectiveness of corridor projects. For future work, it is recommended to expand the implementation of MUSCAT to include more types of sensors. By doing this, the coverage analysis can be further refined and diversified, allowing for more comprehensive assessments.

Additionally, it is crucial to validate the performance of the testbed using real-world data. This validation process will enhance the reliability and applicability of the tool, ensuring its effectiveness in practical scenarios. By addressing these future work areas, MUSCAT can continue to evolve as a powerful tool for sensor placement and coverage analysis in sUAV corridor projects and similar applications.

REFERENCES

- [1] E. Blasch, A. K. Raz, R. Sabatini, and C. C. Insaurralde, "Information fusion as an autonomy enabler for uas traffic management (utm)," in *Proceedings of the AIAA Scitech Forum*, 2021, pp. 1–12.
- [2] A. K. Raz, M. Hieb, J. Ferrari, L. Sherry, and P. Costa, "Exploiting information fusion for cybersecurity of small unmanned aerial vehicles," in *AIAA SCITECH 2023 Forum*, 2023, p. 2582.
- [3] M. Liggins II, D. Hall, and J. Llinas, *Handbook of multisensor data fusion: theory and practice*. CRC press, 2009.
- [4] V. G. Díaz, *Handbook of research on innovations in systems and software engineering*. IGI Global, 2014.
- [5] FAA, "Urban air mobility (uam) concept of operations - version 2.0," 2023.
- [6] S. Verma, V. Dulchinos, R. D. Wood, A. Farrahi, R. Mogford, M. Shyr, and R. Ghata, "Design and analysis of corridors for uam operations," in *2022 IEEE/AIAA 41st Digital Avionics Systems Conference (DASC)*, 2022, pp. 1–10.
- [7] J. Ochodnický, Z. Matousek, M. Babjak, and J. Kurty, "Drone detection by ku-band battlefield radar," in *2017 International Conference on Military Technologies (ICMT)*, 2017, pp. 613–616.
- [8] K. Belwafi, R. Alkadi, S. A. Alameri, H. A. Hamadi, and A. Shoufan, "Unmanned aerial vehicles' remote identification: A tutorial and survey," *IEEE Access*, vol. 10, pp. 87 577–87 601, 2022.
- [9] FAA, "Remote identification of unmanned aircraft - final rule," 2021.
- [10] S. Jovanoska, B. Knoedler, D. P. Palanivelu, L. Still, T. Fiolka, M. Oispuu, C. Steffes, and W. Koch, "Passive sensor processing and data fusion for drone detection," in *Proceedings of the NATO STO Meeting Proceedings: MSG-SET-183 Specialists' Meeting on Drone Detectability: Modelling the Relevant Signature*, Prague, Czech Republic, 2021, p. 16.
- [11] P. Nguyen, H. Truong, M. Ravindranathan, A. Nguyen, R. Han, and T. Vu, "Cost-effective and passive rf-based drone presence detection and characterization," *GetMobile: Mobile Computing and Communications*, vol. 21, no. 4, pp. 30–34, 2018.
- [12] F. Christnacher, S. Hengy, M. Laurenzis, A. Matwyschuk, P. Naz, S. Schertzer, and G. Schmitt, "Optical and acoustical uav detection," in *Electro-Optical Remote Sensing X*, vol. 9988. SPIE, 2016, pp. 83–95.
- [13] I. Guvenc, F. Koohifar, S. Singh, M. L. Sichertiu, and D. Matolak, "Detection, tracking, and interdiction for amateur drones," *IEEE Communications Magazine*, vol. 56, no. 4, pp. 75–81, 2018.
- [14] S. Jovanoska, M. Brötje, and W. Koch, "Multisensor data fusion for uav detection and tracking," in *2018 19th International Radar Symposium (IRS)*. IEEE, 2018, pp. 1–10.
- [15] J. Dudczyk, R. Czyba, and K. Skrzypczyk, "Multi-sensory data fusion in terms of uav detection in 3d space," *Sensors*, vol. 22, no. 12, p. 4323, 2022.
- [16] F. Svanström, F. Alonso-Fernandez, and C. Englund, "A dataset for multi-sensor drone detection," *Data in Brief*, vol. 39, p. 107521, 2021.
- [17] N. Katoh and T. Ibaraki, "Resource allocation problems," *Handbook of Combinatorial Optimization: Volume 1–3*, pp. 905–1006, 1998.
- [18] R. Aievola, F. Causa, G. Fasano, L. Manica, G. Gentile, and M. Dubois, "Ground-based radar networks for urban air mobility: Design considerations and performance analysis," in *2022 IEEE/AIAA 41st Digital Avionics Systems Conference (DASC)*. IEEE, 2022, pp. 1–10.
- [19] "Planning Radar Network Coverage over Terrain example," <https://www.mathworks.com/help/antenna/ug/planning-radar-network-coverage-over-terrain.html>, accessed: 2022-09-30.
- [20] V. Semkin, J. Haarla, T. Pairen, C. Slezak, S. Rangan, V. Viikari, and C. Oestges, "Analyzing radar cross section signatures of diverse drone models at mmwave frequencies," *IEEE access*, vol. 8, pp. 48 958–48 969, 2020.